

HUMAN DEVELOPMENT INDEX (HDI) PREDICTOR

Complete Project Report

Title

Human Development Index (HDI) Predictor - Major Project Report Human Development Index (HDI) Predictor - Major Project Report

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Abstract

This project develops a machine learning-based Human Development Index Predictor. The system estimates HDI values using socio-economic indicators and classifies countries into development categories. The solution combines data preprocessing, machine learning, and a Flask web application. This project develops a machine learning-based Human Development Index Predictor. The system estimates HDI values using socio-economic indicators and classifies countries into development categories. The solution combines data preprocessing, machine learning, and a Flask web application.

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Introduction

The Human Development Index is a composite measure of health, education, and income. Governments and researchers use HDI to evaluate development. This project automates HDI prediction using machine learning. The Human Development Index is a composite measure of health, education, and income. Governments and researchers use HDI to evaluate development. This project automates HDI prediction using machine learning.

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Problem Statement

Manual analysis of development indicators is time-consuming. A predictive system can estimate HDI efficiently and assist policy analysis. Manual analysis of development indicators is time-consuming. A predictive system can estimate HDI efficiently and assist policy analysis.

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Objectives

Predict HDI values, classify countries, build a web application, and provide analytical insights. Predict HDI values, classify countries, build a web application, and provide analytical insights.

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Literature Survey

Previous studies used statistical and machine learning techniques such as Linear Regression, Random Forest, and Neural Networks for socio-economic prediction tasks. Previous studies used statistical and machine learning techniques such as Linear Regression, Random Forest, and Neural Networks for socio-economic prediction tasks.

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System Requirements

Python, Flask, Pandas, NumPy, Scikit-learn, HTML, CSS, Bootstrap, Git, and GitHub. Python, Flask, Pandas, NumPy, Scikit-learn, HTML, CSS, Bootstrap, Git, and GitHub.

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Dataset Description

The dataset contains indicators such as life expectancy, years of schooling, expected years of schooling, and income-related metrics. The dataset contains indicators such as life expectancy, years of schooling, expected years of schooling, and income-related metrics.

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Data Preprocessing

Missing value handling, feature scaling, outlier detection, and dataset splitting were performed before model training. Missing value handling, feature scaling, outlier detection, and dataset splitting were performed before model training.

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Exploratory Data Analysis

Correlation analysis and feature importance evaluation were conducted to understand relationships between indicators and HDI. Correlation analysis and feature importance evaluation were conducted to understand relationships between indicators and HDI.

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Machine Learning Methodology

Linear Regression was selected due to interpretability and strong performance on continuous target prediction. Linear Regression was selected due to interpretability and strong performance on continuous target prediction.

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Model Training

The dataset was divided into training and testing sets. The model was trained using Scikit-learn. The dataset was divided into training and testing sets. The model was trained using Scikit-learn.

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Model Evaluation

Performance was measured using R^2 Score, MAE, MSE, and RMSE metrics. Performance was measured using R^2 Score, MAE, MSE, and RMSE metrics.

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System Architecture

Frontend → Flask Backend → ML Model → Prediction Results. The architecture supports user interaction and prediction generation. Frontend → Flask Backend → ML Model → Prediction Results. The architecture supports user interaction and prediction generation.

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Implementation

Flask handles requests, loads the trained model, processes inputs, and returns predicted HDI values. Flask handles requests, loads the trained model, processes inputs, and returns predicted HDI values.

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User Interface

The application includes input forms, prediction pages, and result visualization components. The application includes input forms, prediction pages, and result visualization components.

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Testing

Functional testing, integration testing, and validation testing were conducted to verify application performance. Functional testing, integration testing, and validation testing were conducted to verify application performance.

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Results

The model successfully predicts HDI values and classifies countries into development categories with satisfactory accuracy. The model successfully predicts HDI values and classifies countries into development categories with satisfactory accuracy.

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Future Enhancements

Future versions may use advanced algorithms, cloud deployment, APIs, and real-time data sources. Future versions may use advanced algorithms, cloud deployment, APIs, and real-time data sources.

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Conclusion

The HDI Predictor demonstrates how machine learning can support development analysis and decision-making processes. The HDI Predictor demonstrates how machine learning can support development analysis and decision-making processes.

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References

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